

20 C resistance of R_3 alone is larger than the effective resistance of R_2 in parallel with R_3

by potential divider rule:

p.d. across R_1 larger

p.d. across R_3 smaller

total circuit resistance drops so battery outputs more power $P = \frac{(\text{emf})^2}{R_{\text{total}}}$

21 A by potential divider rule: